End of Topic 2 Assessment · ANSWER KEY (detailed)

Algebra I · Module 2, Topic 2: Transforming and Comparing Linear Functions · publisher test, Assessments book pp. 19–23 · given Tue Nov 3

Quick key

#	Type	Answer	TEKS	Lesson
1	Fill in	$a = -\frac{1}{2}$	A.3E	Lesson 1 · a·f(x)
2	Multiple choice	**B. $100 \le y \le 250$**	A.2A	Lesson 4 · domain and range in context
3	Multiple choice	**G. $y = 10$; slope is zero**	A.2G	Lesson 3 · horizontal and vertical lines
4	Multiple choice	**D. translated left 1 unit**	A.3E	Lesson 2 · f(x − h)
5	Fill in	**5**	A.3E	Lesson 2 · translating a table
6	Graph	**$g(x) = 3x$: line through (0, 0), (1, 3), (2, 6), (−1, −3)**	A.3E	Lesson 2 · f(bx)
7	Fill in	$y = \frac{3}{2}x - 6$	A.2F	Lesson 3 · perpendicular lines
8	Fill in	$y = 3.2x + 9.6$	A.2E	Lesson 3 · parallel lines
9	Fill in	$y - 2 = \frac{1}{4}x$ or $y = \frac{1}{4}(x + 8)$ or $y - 4 = \frac{1}{4}(x - 8)$	A.2C	Lesson 3 · point-slope form (Topic 1 skill)
10	Select TWO	**H and K**	A.3C	Lesson 4 · key attributes (Topic 1 skill)

Suggested scoring (the publisher gives no point values): 10 points per question = 100. Multiple choice and select-two are all or nothing. Open items (1, 5, 6, 7, 8, 9): full credit for the answer in any equivalent form; half credit as described under each item. TEKS on this test (TE p. 144D): A.2A, A.2C, A.2E, A.2F, A.2G, A.3C, A.3E. Readiness standards: A.2C, A.3C.

Question 1 · Fill in

The graph of m passes through (−6, 3) and (4, −2). Given f(x) = x and m(x) = af(x), find a.

$a = -\frac{1}{2}$

Solution

1. $m(x) = a \cdot f(x) = ax$, so a is the **slope** of the line (the line goes through the origin).
2. slope $= \frac{-2-3}{4-(-6)} = \frac{-5}{10} = -\frac{1}{2}$
3. So $a = -\frac{1}{2}$ (also accept −0.5).

Check

Plug in a point: −½(−6) = 3 ✓ and −½(4) = −2 ✓. The line falls left to right, so a must be negative ✓.

If a student chose / wrote…

−2	Used run ÷ rise (10 ÷ −5) instead of rise ÷ run.
½ or 0.5	Sign error: subtracted y-values in one order and x-values in the other.
−5 or 10	Stopped at the rise or the run; didn't divide.
3 or −6	Read a coordinate of a labeled point instead of finding the slope.

Partial credit

2 pts: −½. 1 pt: correct slope setup with one arithmetic or sign slip, or ½ with work that shows rise/run.

Re-teach (Wed Nov 4 retakes)

Slope triangle on the grid between the two labeled points; say "a stretches f(x) = x, so a IS the slope."

Question 2 · Multiple choice

Lawn service: $100 fixed + $25 per hour, at most 6 hours. What is the range for one day?

B. $100 \le y \le 250$

Solution

1. Range = the possible **costs** (*y*-values). Domain = the hours (*x*-values).
2. Least cost: 0 hours → 100 + 25(0) = $100.
3. Greatest cost: 6 hours → 100 + 25(6) = 100 + 150 = $250.
4. Range: $100 \le y \le 250$ → **B**.

Check

Two choices (C, D) are about x, so they describe domain, not range. Of A and B, only B starts at the $100 fixed fee.

If a student chose / wrote…

A. $25 \le y \le 400$	Started at the $25 hourly rate instead of the $100 fixed fee; the maximum is too high (6 hours costs $250).
C. $0 \le x \le 6$	This is the **domain** (hours). Mixed up domain and range.
D. $0 \le x \le 8$	Domain idea with the wrong maximum (8-hour work day assumption).

Partial credit

Multiple choice: all or nothing.

Re-teach (Wed Nov 4 retakes)

Two-column chart: Domain = inputs = x = hours | Range = outputs = y = dollars. Evaluate at both ends.

Question 3 · Multiple choice

What are the equation and slope of the horizontal line through $y = 10$?

G. $y = 10$; slope is zero

Solution

1. Every point on the line has y-value 10: (−40, 10), (0, 10), (40, 10). So the equation is $y = 10$.
2. Rise between any two points = 10 − 10 = 0, so slope = 0 ÷ run = **0**.
3. Answer **G**.

Check

Horizontal = "y = number, slope zero". Vertical = "x = number, slope undefined".

If a student chose / wrote…

F. x = 10; slope is 10	Wrong variable and treated the 10 as the slope.
H. x = 10; slope is zero	Right slope, but named the line by x (that would be a vertical line).
J. y = 10; slope is undefined	Right equation, but mixed up zero (horizontal) and undefined (vertical) slope.

Partial credit

Multiple choice: all or nothing.

Re-teach (Wed Nov 4 retakes)

"HOY / VUX": Horizontal, 0 slope, y = · Vertical, Undefined slope, x = .

Question 4 · Multiple choice

Catalina graphed f(x) and g(x) = f(x + 1). How are the graphs related?

D. translated left 1 unit

Solution

1. A change **inside** the parentheses moves the graph sideways.
2. $f(x + 1) = f(x - (-1))$, so $h = -1$: the graph moves **left** 1.
3. Test a point: g reaches f's value at $x = 0$ when $x = -1$, one unit to the left. Answer **D**.

Check

Make a quick table with f(x) = x: g(x) = x + 1 goes through (−1, 0), f goes through (0, 0): left 1 ✓ (it is also up 1 for this f, but the general rule is left).

If a student chose / wrote…

A. up 1	Treated + 1 inside like + 1 outside (vertical shift).
B. down 1	Vertical shift and wrong direction.
C. right 1	Followed the sign: + → right. Inside the parentheses the direction is opposite.

Partial credit

Multiple choice: all or nothing.

Re-teach (Wed Nov 4 retakes)

"Inside lies": inside the parentheses the graph moves opposite the sign. Desmos slider on h.

Question 5 · Fill in

Table for f: (−2, 7), (−1, 5), (0, 3), (1, 1), (2, −1), (3, −3). Translate f 1 unit right. What is the new y-intercept?

5

Solution

1. Right 1 means add 1 to every x; the y-values stay the same.
2. The new y-intercept is where the new x is 0. The old point with x = −1 is (−1, 5); it moves to (0, 5).
3. Or by rule: $f(x) = -2x + 3$, so $g(x) = f(x - 1) = -2(x - 1) + 3 = -2x + 5$. y-intercept **5**.

Check

New table: (−1, 7), (0, 5), (1, 3)… still drops by 2 each step ✓.

If a student chose / wrote…

1	Moved the wrong way (left), so the point (1, 1) landed at (0, 1).
3	Gave the original y-intercept; didn't translate.
4	Added 1 to the y-intercept (treated "right" as "up").
(0, 5)	Correct; accept the point form.

Partial credit

2 pts: 5 or (0, 5). 1 pt: correct new table or g(x) = −2x + 5 but wrong intercept read off.

Re-teach (Wed Nov 4 retakes)

Have students rewrite the table with x + 1 in a new column, then find the row where x = 0.

Question 6 · Graph

f(x) = x is graphed. Graph g(x) = f(3x).

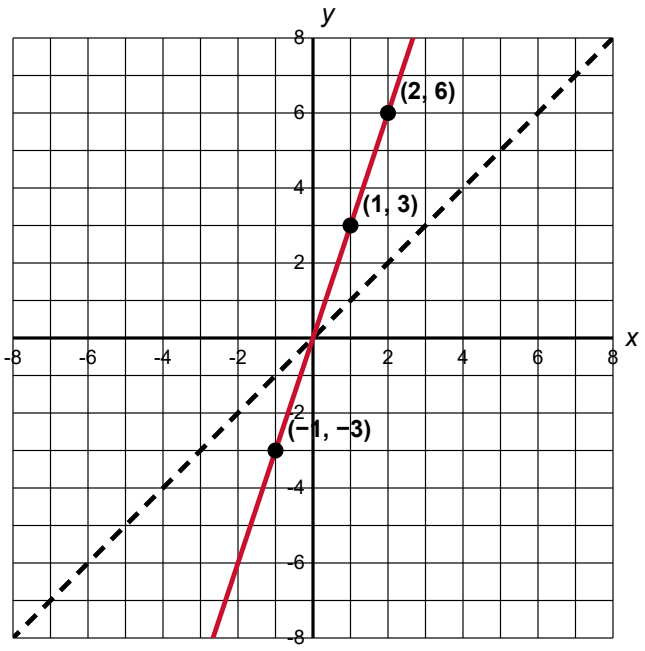
g(x) = 3x: line through (0, 0), (1, 3), (2, 6), (−1, −3)

Solution

1. $f(x) = x$ means "the output equals the input". $f(3x)$ means: put **3*x*** into *f* instead of *x*.
2. So $g(x) = f(3x) = 3x$. This is a line with **slope 3** and **y-intercept 0**.
3. Make a table. For each x, first multiply by 3, then that is the output:

x	*f*(*x*) = *x*	3*x*	*g*(*x*) = *f*(3*x*)	Point to plot
−2	−2	3(−2) = −6	**−6**	(−2, −6)
−1	−1	3(−1) = −3	**−3**	(−1, −3)
0	0	3(0) = 0	**0**	(0, 0)
1	1	3(1) = 3	**3**	(1, 3)
2	2	3(2) = 6	**6**	(2, 6)

4. Plot the five points from the last column: (−2, −6), (−1, −3), (0, 0), (1, 3), (2, 6).
5. Connect them with a straight line and put arrows on both ends. Label it g(x).
6. Compare with f: f rises 1 for every 1 across; g rises **3** for every 1 across, so g is 3 times as steep. Both pass through (0, 0).



Check

g should be 3 times as steep as f and still pass through (0, 0). At x = 2 it reaches 6, f only reaches 2.

If a student chose / wrote…

Line with slope ⅓	Thought f(3x) makes the graph wider by 3; for f(x) = x the graph gets steeper.
y = x + 3 (shifted up 3)	Treated 3x as + 3.
y = x shifted left 3	Treated f(3x) like f(x + 3).

Partial credit

2 pts: correct line through the origin with slope 3 (at least two correct points). 1 pt: correct table or equation g(x) = 3x, graph off.

Re-teach (Wed Nov 4 retakes)

Table first, graph second. Compare f(2x), f(3x), f(½x) on one grid.

Question 7 · Fill in

A.2F · Lesson 3 · perpendicular lines

Line crosses the x-axis at 4 and is perpendicular to $y = -\frac{2}{3}x + 4$. Write it in slope-intercept form.

$$\boxed{y = \frac{3}{2}x - 6}$$

Solution

1. Given slope $-\frac{2}{3}$. Perpendicular slope = negative reciprocal = $\mathbf{\frac{3}{2}}$.
2. "Crosses the x-axis at 4" is the point (4, 0).
3. $0 = \frac{3}{2}(4) + b \rightarrow 0 = 6 + b \rightarrow b = -6$.
4. $y = \frac{3}{2}x - 6$ (also accept $y = 1.5x - 6$).

Check

Slopes multiply to −1: $-\frac{2}{3} \times \frac{3}{2} = -1$ ✓. Point: $\frac{3}{2}(4) - 6 = 0$ ✓.

If a student chose / wrote…

y = −⅔x + 8/3	Used the same slope (parallel, not perpendicular).
y = −3/2x + 6	Took the reciprocal but kept the negative sign.
y = 3/2x + 4	Kept the original y-intercept instead of using (4, 0).
y = 3/2x − 4 or point (0, 4)	Read "crosses the x-axis at 4" as the y-intercept.

Partial credit

2 pts: y = 3/2x − 6. 1 pt: correct slope 3/2 with a wrong b, or correct b from a wrong slope with work shown.

Re-teach (Wed Nov 4 retakes)

Flip-and-switch drill on whiteboards (2/3 → −3/2), then "x-axis at 4 means (4, 0)".

Question 8 · Fill in

Line through (−3, 0) parallel to y = 3.2x + 7.3. Write it in slope-intercept form.

$y = 3.2x + 9.6$

Solution

1. Parallel lines have the **same slope**: 3.2.
2. $0 = 3.2(-3) + b \rightarrow 0 = -9.6 + b \rightarrow b = 9.6$.
3. $y = 3.2x + 9.6$

Check

$3.2(-3) + 9.6 = -9.6 + 9.6 = 0$ ✓.

If a student chose / wrote…

y = 3.2x + 7.3	Copied the given line; didn't use the point.
y = 3.2x − 9.6	Sign error solving for b (0 = −9.6 + b).
y = −0.3125x + …	Used the perpendicular slope (−1/3.2).
y − 0 = 3.2(x + 3)	Correct line in point-slope form: give 1 pt (question asks for slope-intercept).

Partial credit

2 pts: y = 3.2x + 9.6. 1 pt: correct point-slope form, or correct slope with an arithmetic slip in b.

Re-teach (Wed Nov 4 retakes)

Same slope, new point, solve for b: three-step frame on an index card.

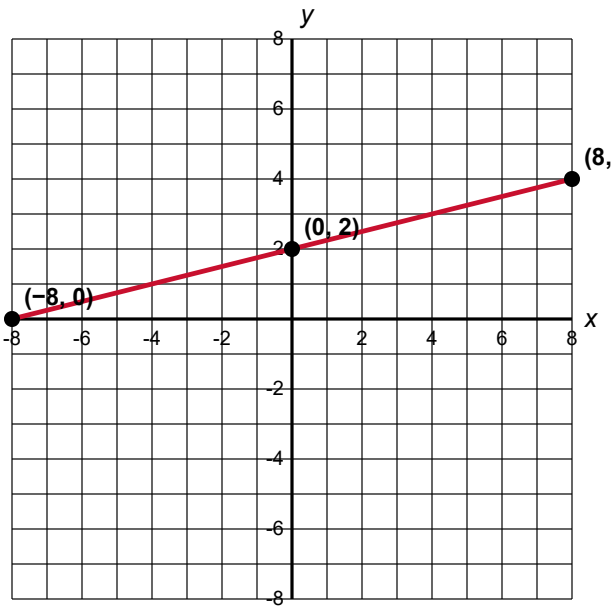
Question 9 · Fill in

Write an equation in point-slope form for the graphed line.

$y - 2 = \frac{1}{4}x$ or $y = \frac{1}{4}(x + 8)$ or $y - 4 = \frac{1}{4}(x - 8)$

Solution

1. Read two exact grid points: (−8, 0) and (0, 2). (The line also passes through (8, 4).)
2. slope = $\frac{2 - 0}{0 - (-8)} = \frac{2}{8} = \frac{1}{4}$
3. Point-slope form $y - y_1 = m(x - x_1)$ with (−8, 0): $y - 0 = \frac{1}{4}(x + 8)$.
4. Any point on the line is correct: (0, 2) gives y − 2 = ¼(x − 0); (8, 4) gives y − 4 = ¼(x − 8).



Check

All three forms simplify to $y = \frac{1}{4}x + 2$. Rises 1 for every 4 across on the grid ✓.

If a student chose / wrote…

slope 4	Run ÷ rise.
y − 0 = ¼(x − 8)	Sign error with the point (−8, 0): x − (−8) = x + 8.
y = ¼x + 2	Correct line, but slope-intercept form, not point-slope: 1 pt.
slope −¼	Line rises left to right; slope must be positive.

Partial credit

2 pts: any correct point-slope equation. 1 pt: correct slope with a sign slip on the point, or the correct line in another form.

Re-teach (Wed Nov 4 retakes)

Circle two lattice points first, slope triangle, then fill y − __ = __(x − __).

Question 10 · Select TWO

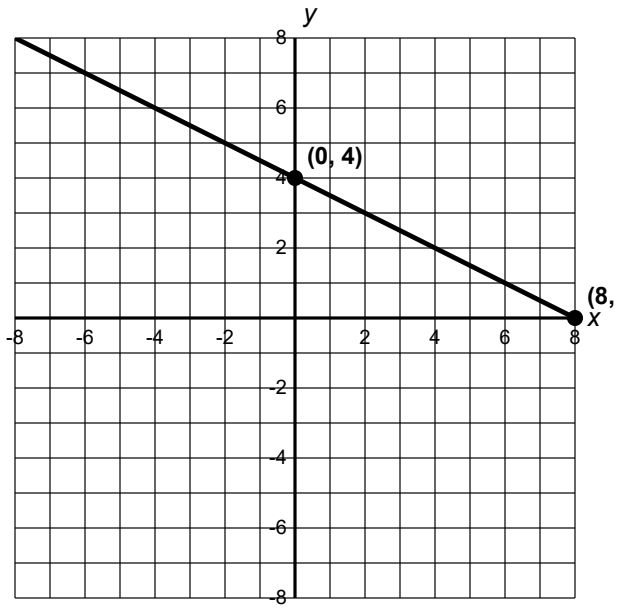
A.3C · Lesson 4 · key attributes (Topic 1 skill)

Line through (0, 4) and (8, 0). Which TWO statements describe it?

H and K

Solution

1. The line crosses the y-axis at (0, 4) → **H is true**.
2. It crosses the x-axis at (8, 0), so the zero is 8 → **K is true**.
3. slope = $\frac{0-4}{8-0} = -\frac{1}{2}$: negative, so the graph is decreasing (F and G are false).
4. J is false: 4 is the y-intercept, not the zero.



Check

Exactly two answers. Falls left to right ✓ (decreasing); y = −½x + 4 → y = 0 when x = 8 ✓.

If a student chose / wrote…

F. increasing	Read the line right to left.
G. slope ½	Dropped the negative sign.
J. zero is 4	Mixed up the zero (x-intercept) and the y-intercept.

Partial credit

STAAR scores select-two items all or nothing (both H and K). Optional classroom partial credit: 1 of 2 pts for one correct, no wrong.

Re-teach (Wed Nov 4 retakes)

"Zero = where y is zero = x-intercept". Hands-on: trace the line left to right for increasing/decreasing.

Item analysis and retake groups

Tally the number of students who missed each question. A question missed by 30% or more = whole-class re-teach on Nov 4; otherwise small group.

#	1	2	3	4	5	6	7	8	9	10
Answer	**−½**	**B**	**G**	**D**	**5**	**y = 3x**	**y = 1.5x − 6**	**y = 3.2x + 9.6**	**y = ¼(x + 8)**	**H, K**
Per. 1										
Per. 2										
Per. 3										
Per. 4										
Per. 5										
Per. 6										
Per. 7										
% missed										

Retake groups by skill

Group	Questions	Re-teach focus	Students
Transformations (A.3E)	1, 4, 5, 6	Inside vs. outside; a = slope; table shifts	
Domain and range (A.2A)	2	Inputs vs outputs in context, evaluate at both ends	
Horizontal / vertical lines (A.2G)	3	HOY / VUX	
Parallel and perpendicular (A.2E, A.2F)	7, 8	Same slope / flip-and-switch, then solve for b	
Writing equations from a graph (A.2C)	9	Two lattice points, slope triangle, point-slope frame	
Key attributes (A.3C)	10	Zero vs y-intercept; increasing vs decreasing	

Retake form + key: Module 2 topic 2/Topic2-Retake(-ANSWER-KEY).pptx (IEP and 504 versions in IEP/ and Section 504/).